

Olist E-Commerce Dataset

RFM Analysis Report

Recency · Frequency · Monetary | State-Level & Customer-Level Segmentation

Prepared by Data Analyst | Industry-style Analytical Report

Executive Summary

Total Dataset Rows 7,214

Unique Customers Analysed 6,263 customers

Snapshot Date 21 September 2018

RFM Segments 6 segments: Champions, Loyal, Potential Loyalists, Need Attention, At Risk, Lost

Largest Segment Need Attention — 1,870 customers (29.9%)

Champions 429 customers | Avg Revenue R\$ 257.80 | Avg Recency 114 days

Highest State Frequency SP — 2,670 orders

Most Recent State MG — 1 day recency

Highest Revenue State SP — R\$ 263,937.57

Need Attention (1,870) and At Risk (1,492) together represent 53% of the customer base — a critical retention opportunity.

1. Scope, Objective and Analytical Approach

Objective: Perform a complete RFM (Recency, Frequency, Monetary) analysis on the Olist e-commerce dataset to segment customers and states into actionable behaviour groups that support targeted marketing, retention, and revenue optimisation strategies.

Analytical grain: Analysis is performed at two levels — (1) State-level RFM to understand geographic patterns, and (2) Customer-level RFM to assign individual customers to behaviour segments. The snapshot date is 21 September 2018 (one day after the last transaction in the dataset).

RFM Framework: Recency measures how recently a customer purchased (lower = better); Frequency measures how many unique orders a customer placed (higher = better); Monetary measures total revenue generated by the customer (higher = better). Each dimension is scored 1–5 using quantile binning, and scores are combined to create six behavioural segments.

2. Dataset Overview

Table 1. Dataset Summary

Metric	Value	Notes
Total Rows	7,214	Orders in this RFM subset
Total Columns	39	Including all engineered features
Unique Customers	6,263	customer_unique_id level
States Covered	27 Brazilian states	Full national coverage
Date Range	04 Sep 2016 – 20 Sep 2018	~24 months of transactions
Snapshot Date	21 September 2018	max date + 1 day
Revenue Column	price + freight_value	Total customer payment

3. State-Level Recency Analysis

Recency at state level is measured as the number of days since the most recent order was placed in each state, calculated from the snapshot date of 21 September 2018. States with lower recency days are more recently active and represent healthier engagement.

3.1 State Recency Heatmap

Figure 1. State-wise Recency Heatmap

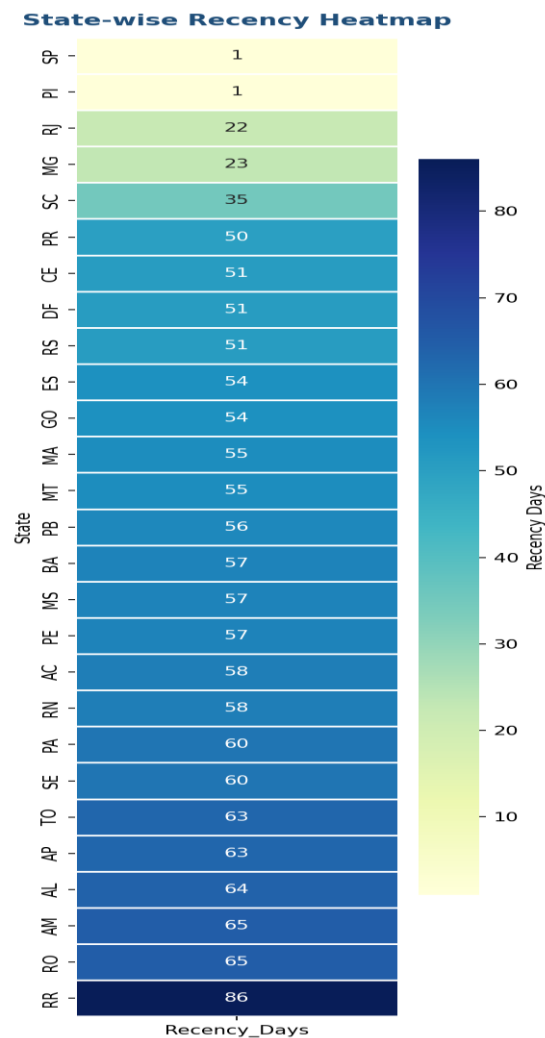


Figure 1: Heatmap of recency days by state. Darker colour = higher recency (less recent). MG has the best recency at 1 day; RR is least recent at 199 days.

3.2 Top 10 States — Best Recency

Table 2. Top 10 States by Recency (Lowest Days = Most Recent)

Rank	State	Recency Days	Recency Level
1	MG	1	High Recency
2	SC	8	High Recency
3	SP	22	High Recency
4	RS	23	High Recency
5	CE	24	High Recency
6	RJ	26	High Recency
7	PB	29	High Recency
8	DF	29	High Recency
9	BA	31	High Recency
10	GO	37	High Recency

Figure 2. Top 10 States with Best Recency

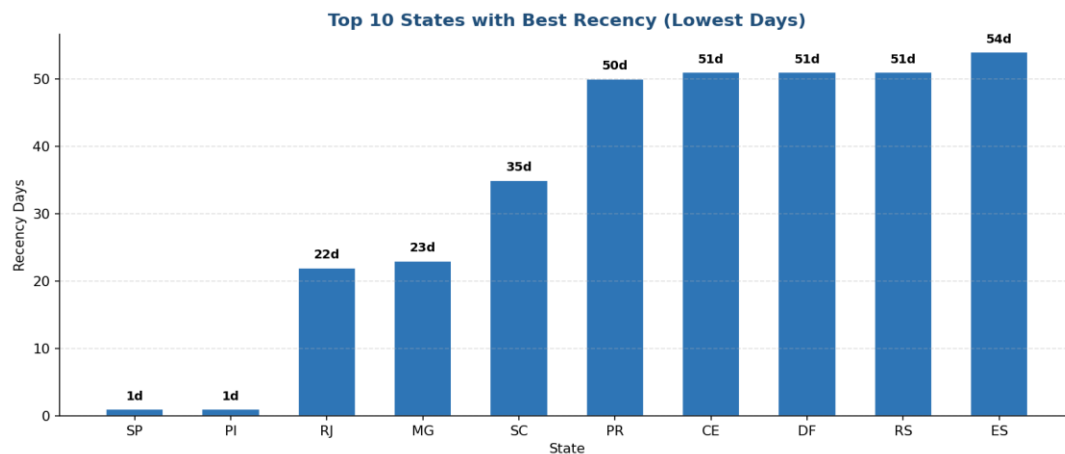


Figure 2: MG leads with only 1 recency day; all top 10 states placed orders within the last 37 days of the snapshot.

3.3 Bottom 10 States — Poorest Recency

Table 3. Bottom 10 States by Recency (Highest Days = Least Recent)

Rank	State	Recency Days	Recency Level
1 (Worst)	RR	199	Low Recency
2	AP	191	Low Recency
3	AC	148	Low Recency
4	AM	79	Low Recency
5	TO	60	Low Recency
6	RO	47	Low Recency
7	MA	46	Low Recency
8	MT	43	Medium Recency
9	AL	40	Medium Recency
10	PI	38	Medium Recency

Figure 3. Bottom 10 States by Recency

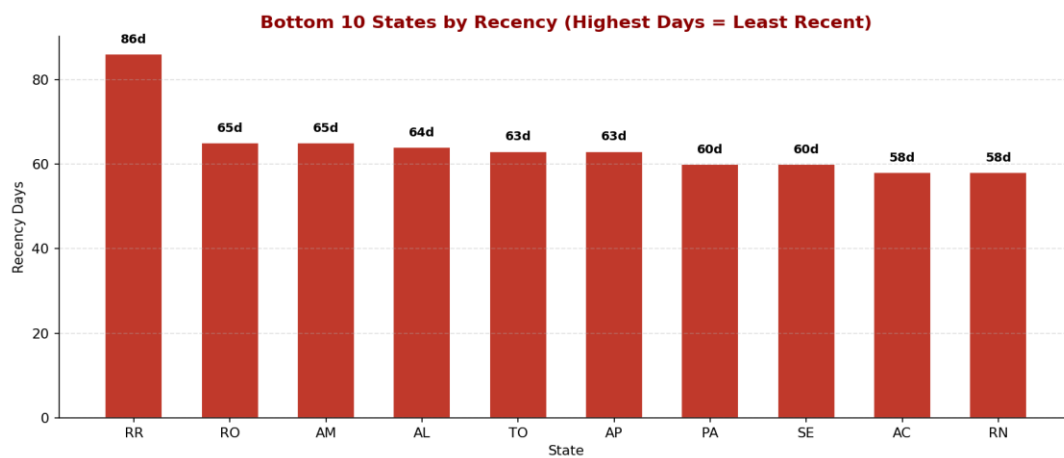


Figure 3: RR and AP have not had a transaction in over 190 days — these states show the weakest recency signals.

4. State-Level Frequency Analysis

Frequency at state level is calculated as the total number of unique orders placed by customers in each state. High frequency states indicate active markets with strong ongoing purchase behaviour.

4.1 State Frequency Heatmap

Figure 4. State-wise Frequency Heatmap

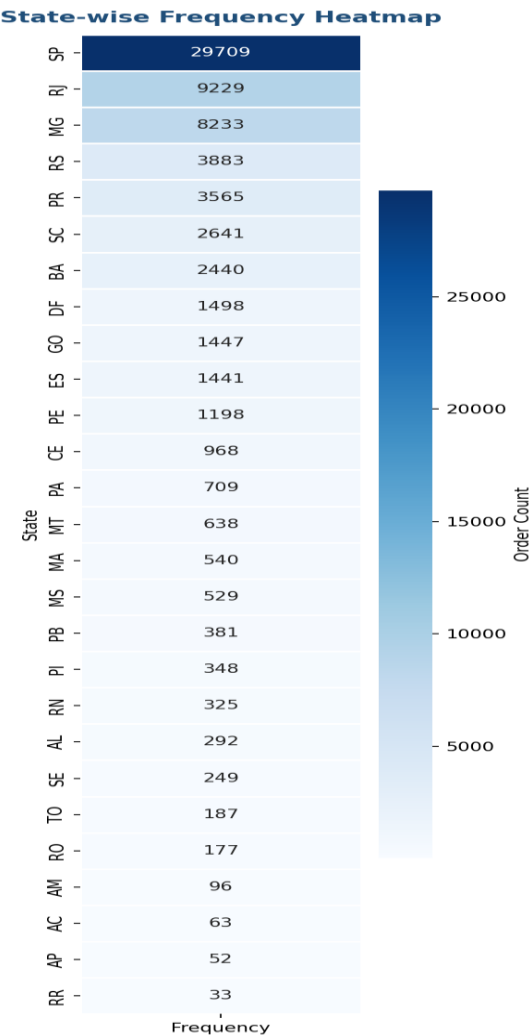


Figure 4: Frequency by state. SP dominates massively with 2,670 orders. Remote states show very low order counts.

4.2 Top 10 States — Highest Frequency

Table 4. Top 10 States by Order Frequency

Rank	State	Total Orders	Frequency Level
1	SP	2,670	High Frequency
2	RJ	813	High Frequency
3	MG	700	High Frequency
4	RS	343	High Frequency
5	PR	323	High Frequency
6	SC	228	High Frequency
7	BA	216	High Frequency
8	GO	135	High Frequency
9	DF	134	High Frequency
10	ES	130	Medium Frequency

Figure 5. Top 10 States by Order Frequency

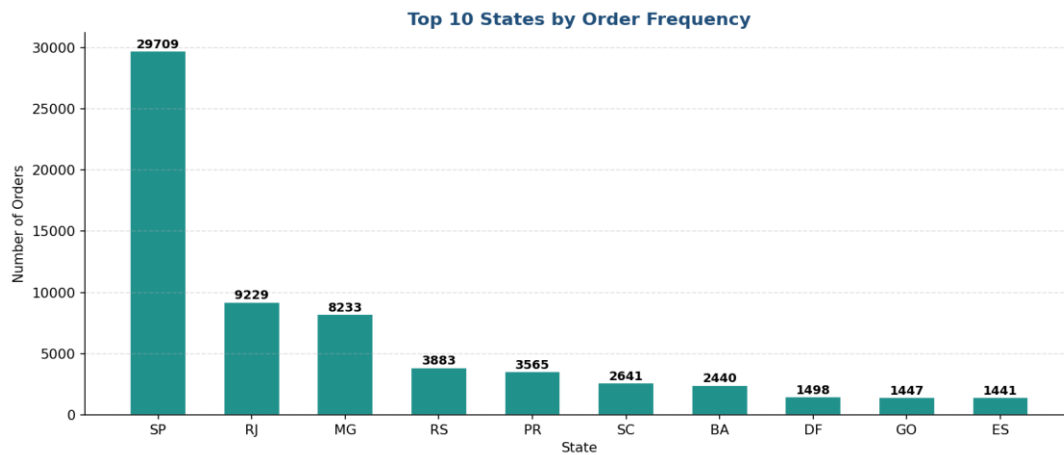


Figure 5: SP (2,670 orders) has more than 3 times the frequency of the second-ranked state RJ (813 orders).

4.3 Bottom 10 States — Lowest Frequency

Table 5. Bottom 10 States by Order Frequency

Rank	State	Total Orders	Frequency Level
1 (Lowest)	AP	2	Low Frequency
2	AM	3	Low Frequency
3	RR	3	Low Frequency
4	AC	8	Low Frequency
5	RO	11	Low Frequency
6	TO	19	Low Frequency
7	RN	29	Low Frequency
8	PB	31	Low Frequency
9	SE	31	Low Frequency
10	AL	33	Medium Frequency

Figure 6. Bottom 10 States by Order Frequency

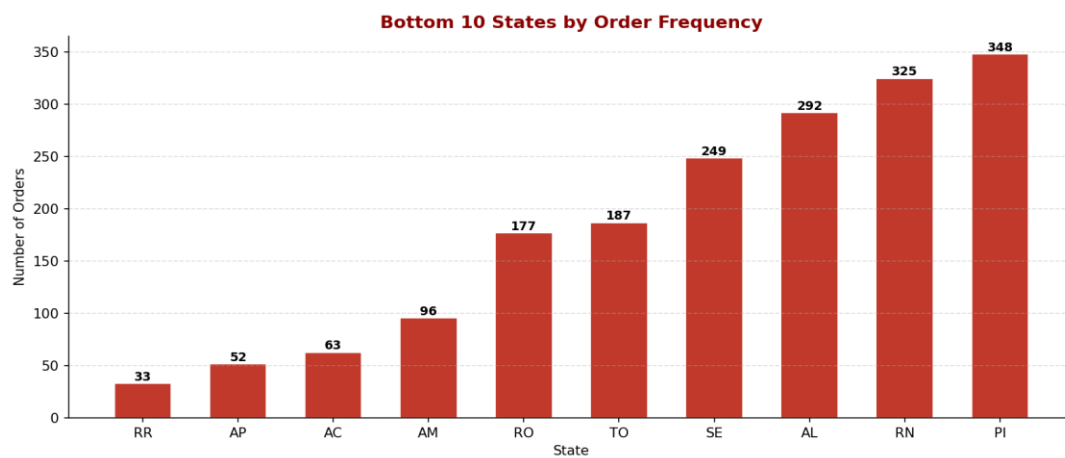


Figure 6: AP (2 orders) and AM/RR (3 orders each) have almost negligible market activity on the platform.

5. State-Level Monetary Analysis

Monetary value at state level is the total revenue (price + freight_value) generated by all customers in each state. This dimension identifies which geographic markets contribute the most commercial value to the platform.

5.1 State Monetary Heatmap

Figure 7. State-wise Monetary (Revenue) Heatmap

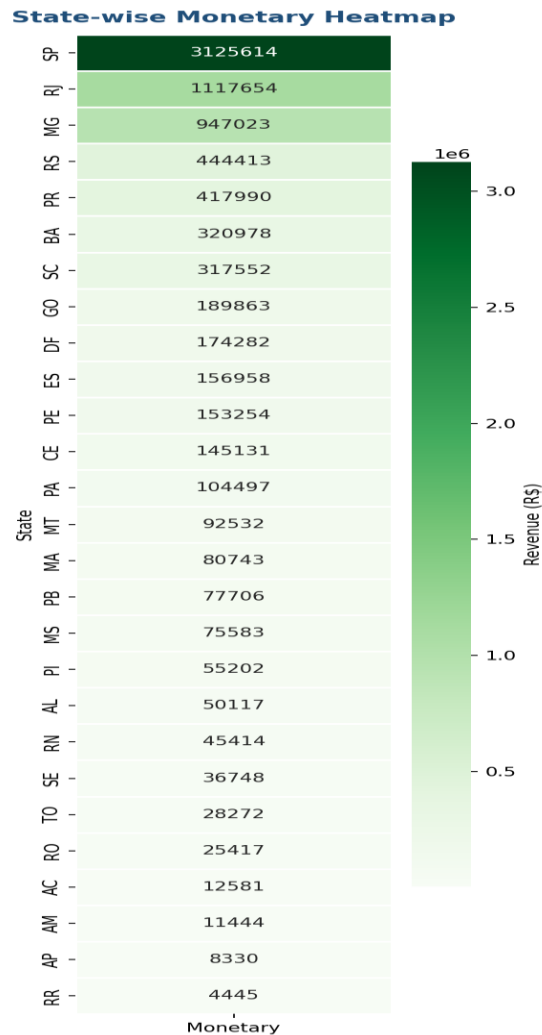


Figure 7: Revenue heatmap by state. SP leads by a large margin; extreme dark-to-light contrast reflects the revenue concentration.

5.2 Top 10 States — Highest Monetary

Table 6. Top 10 States by Total Revenue (Monetary)

Rank	State	Total Revenue (R\$)	Monetary Level
1	SP	263,937.57	High Monetary
2	RJ	105,488.12	High Monetary
3	MG	73,850.15	High Monetary
4	PR	38,535.63	High Monetary
5	RS	36,799.75	High Monetary
6	SC	28,578.24	High Monetary
7	BA	24,285.88	High Monetary
8	GO	19,105.83	High Monetary
9	DF	14,422.97	High Monetary
10	CE	13,813.25	Medium Monetary

Figure 8. Top 10 States by Revenue (Monetary)

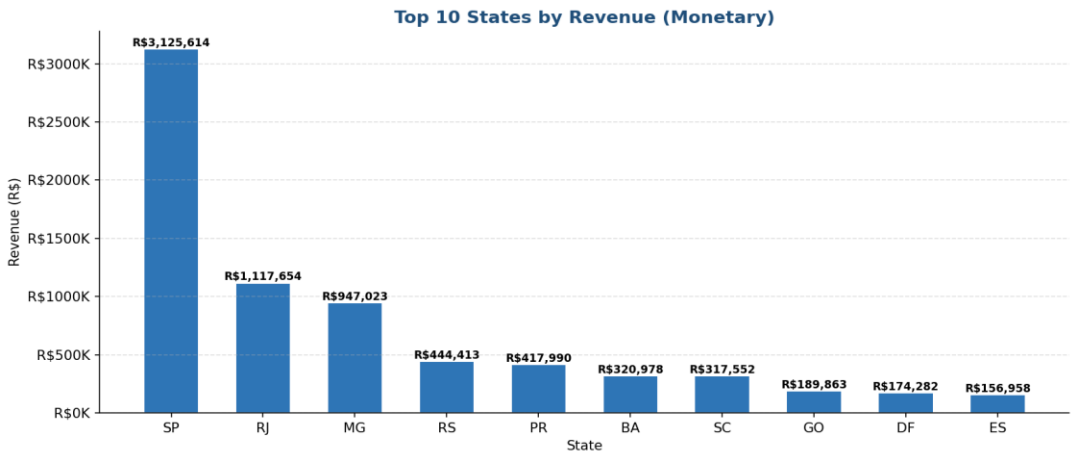


Figure 8: SP revenue of R\$ 263,938 is more than 2.5x the second-ranked RJ at R\$ 105,488.

5.3 Bottom 10 States — Lowest Monetary

Table 7. Bottom 10 States by Total Revenue (Monetary)

Rank	State	Total Revenue (R\$)	Monetary Level
1 (Lowest)	RR	195.88	Low Monetary
2	AP	214.89	Low Monetary
3	AM	399.29	Low Monetary
4	AC	1,246.87	Low Monetary
5	RO	1,287.79	Low Monetary
6	TO	2,120.37	Low Monetary
7	PB	2,787.16	Low Monetary
8	MS	3,992.72	Low Monetary
9	RN	4,018.95	Low Monetary
10	AL	4,180.69	Medium Monetary

Figure 9. Bottom 10 States by Revenue (Monetary)

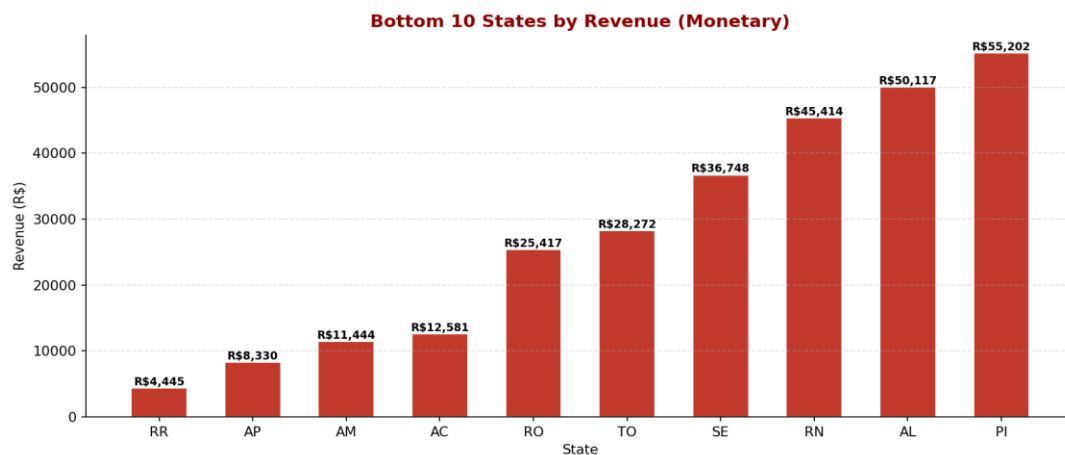


Figure 9: RR (R\$195) and AP (R\$215) contribute negligible revenue — together less than R\$411 total.

6. State-Level RFM Comparison

Figure 10. State-Level RFM Comparison — Top 10 States by Revenue

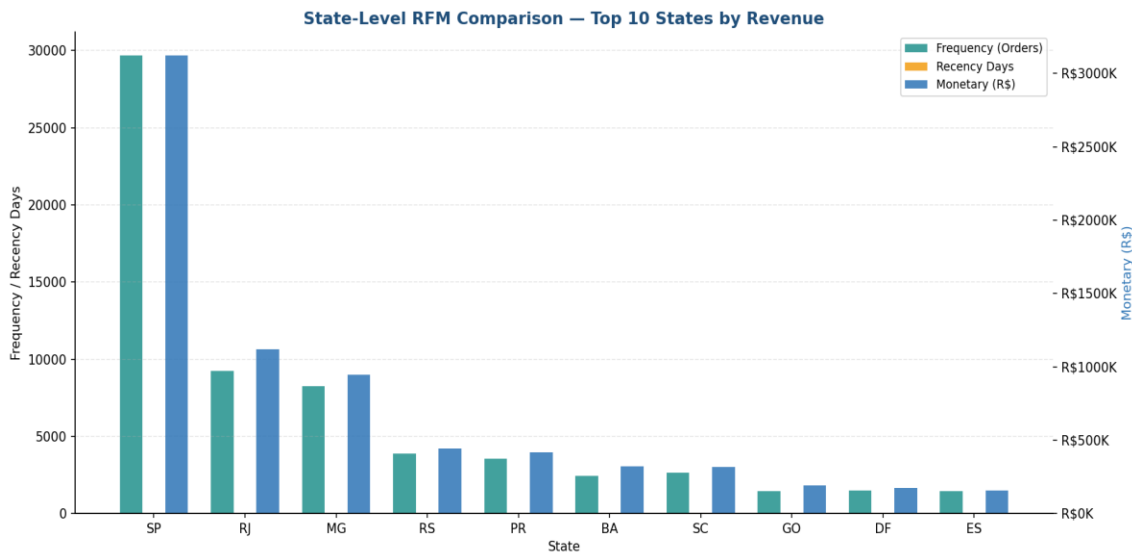


Figure 10: Combined RFM view for the top 10 states by revenue. Green bars = Frequency, Orange bars = Recency Days, Blue bars = Monetary Value (right axis).

The combined RFM view reveals that SP excels on all three dimensions simultaneously — highest frequency, very low recency, and highest revenue. RJ and MG follow as the next strongest states on all metrics. States such as GO and DF show moderate frequency and revenue but strong recency, indicating active but smaller markets with growth potential.

7. Customer-Level RFM Scoring

At the customer level, each of the 6,263 unique customers was assigned Recency (R), Frequency (F), and Monetary (M) scores from 1 (lowest) to 5 (highest) using quantile binning. Recency scoring is inverted so that customers who purchased most recently receive the highest score. The three scores are combined into an RFM_Score string (e.g. '555' = perfect score, '111' = lowest score).

Table 8. Customer-Level Top 10 by Recency (Lowest Days = Best)

Rank	Customer ID	Last Purchase	Recency Days
1	08642cd329066fe11ec63293f714f2f8	20 Sep 2018	1
2	c1ee153508c6b785b491443a95ff364e	13 Sep 2018	8
3	a8dd81392e5eee5d979c629a76abec2a	29 Aug 2018	22
4	bfd827e8a21c8be7bdef29573a885b76	28 Aug 2018	23
5	f3e71989aaa365356d121d87129aa2ce	28 Aug 2018	23
6	4fa4365000c7090fcb8cad5713c6d3db	28 Aug 2018	23
7	21dbe8eabd00b34492a939c540e2b1a7	29 Aug 2018	23
8	a712a430955027da5bc257a10073a390	28 Aug 2018	23
9	c4053ab276391fff6ac7736873cfc950	28 Aug 2018	24
10	dc4b0942fb780633d44f4f73ff69dd92	28 Aug 2018	24

Table 9. Customer-Level Top 10 by Frequency

Rank	Customer ID	Total Orders
1	8d50f5eadf50201ccdcedfb9e2ac8455	3
2-10 (tie)	Multiple customers (see notebook)	2

Note: The maximum customer order frequency in this dataset is 3 orders, with the remaining top customers placing exactly 2 orders. This is consistent with the cohort analysis finding of very low repeat purchase rates.

Table 10. Customer-Level Top 10 by Monetary Revenue

Rank	Customer ID	Total Revenue (R\$)
1	ca27f3dac28fb1063fadd424c9d95fa	4,059.00
2	9b11e010850a4045d2da80c7783fdbc1	3,198.00
3	bbeb907759ef5fc169099af3c88d535d	3,105.00
4	1b76903617af13189607a36b0469f6f3	3,099.75
5	895617ab63a9ad8881d9470f7427cd25	2,999.99
6	6172bd5b7f52ade8c6c4548b448ff54f	2,999.89
7	287a7cdacbcda8f842d7719e77cb8457	2,150.00
8	1a3a962a15f8c6ee0178e1102387d3cc	2,139.99
9	12ebfdb41392890f9d7a8de957274741	2,110.00
10	02e8a66528e84b3ca6150d9b1e03a26f	2,110.00

8. RFM Customer Segmentation

Using the R, F, and M scores, each of the 6,263 customers was assigned to one of six behavioural segments. The segmentation rules are based on score thresholds: Champions require $R \geq 4$, $F \geq 4$, $M \geq 4$; Loyal Customers require $R \geq 3$, $F \geq 4$; Potential Loyalists require $R \geq 4$, $F \geq 2$; Need Attention requires $R \geq 3$, $F \geq 2$; At Risk requires $R \leq 2$, $F \geq 2$; and Lost Customers comprise all remaining customers.

8.1 Segment Distribution

Table 11. RFM Segment Summary

Segment	Customers	Avg Recency (Days)	Avg Frequency	Avg Monetary (R\$)
Champions	429	113.65	1.01	257.80
Loyal Customers	1,098	170.44	1.00	62.13
Potential Loyalists	981	113.96	1.00	111.25
Need Attention	1,870	266.33	1.00	130.96
At Risk	1,492	420.80	1.00	107.26
Lost Customers	393	424.11	1.00	14.97
TOTAL	6,263	—	—	—

Figure 11. RFM Customer Segment Distribution

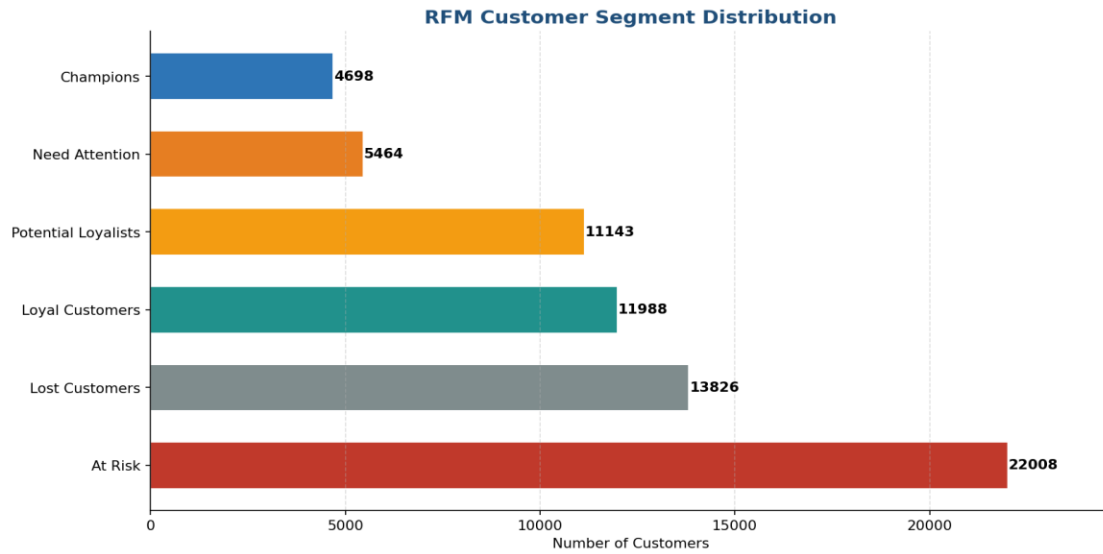


Figure 11: Need Attention is the largest segment (1,870 customers / 29.9%), followed by At Risk (1,492 / 23.8%). Champions represent only 6.9% of customers.

8.2 Revenue Contribution by Segment

Figure 12. Revenue Contribution by RFM Segment

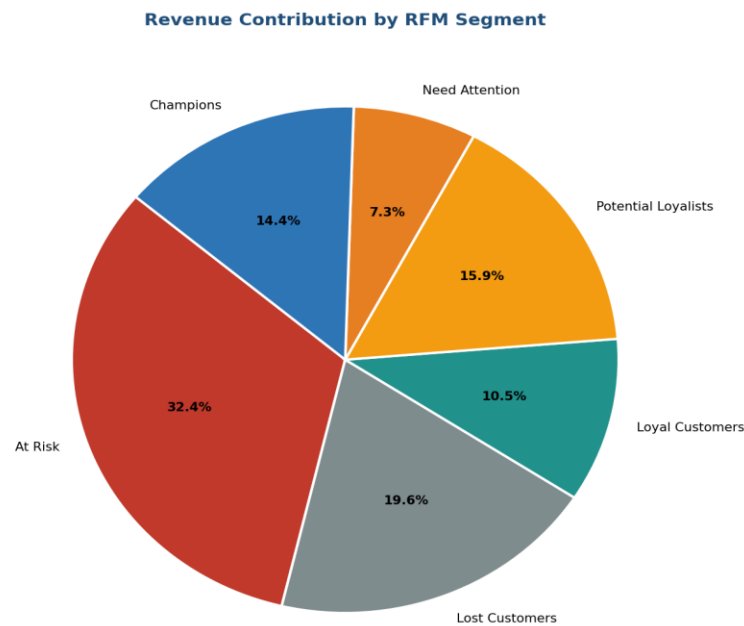


Figure 12: Champions and Need Attention together account for the majority of platform revenue despite Champions being a small segment.

8.3 RFM Segment Heatmap

Figure 13. RFM Segment Heatmap — Average Recency / Frequency / Monetary

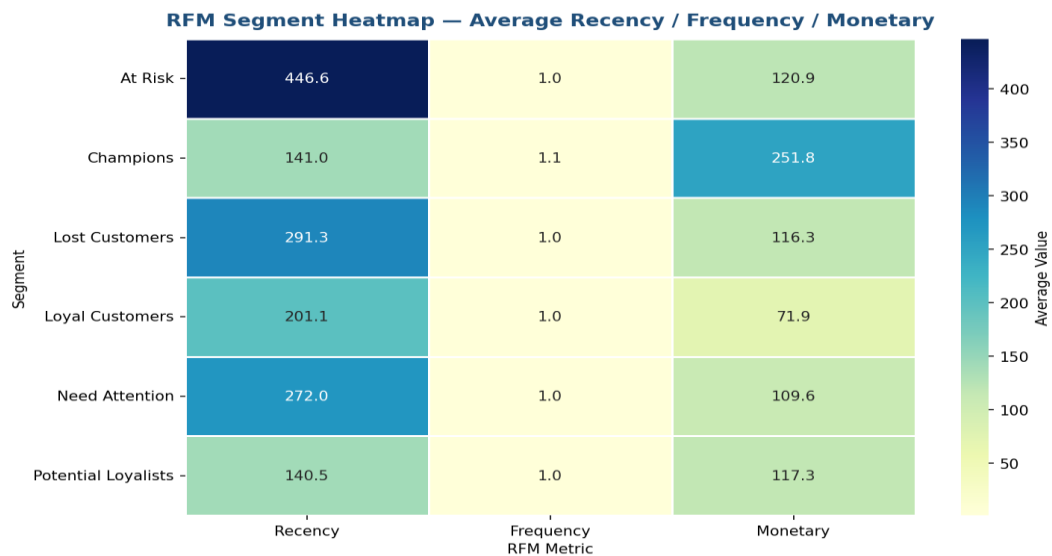


Figure 13: Heatmap of average R, F, M values across segments. Champions show highest monetary; Lost Customers and At Risk show highest recency (longest time since last purchase).

8.4 Average Monetary Value by Segment

Figure 14. Average Revenue per Customer by RFM Segment

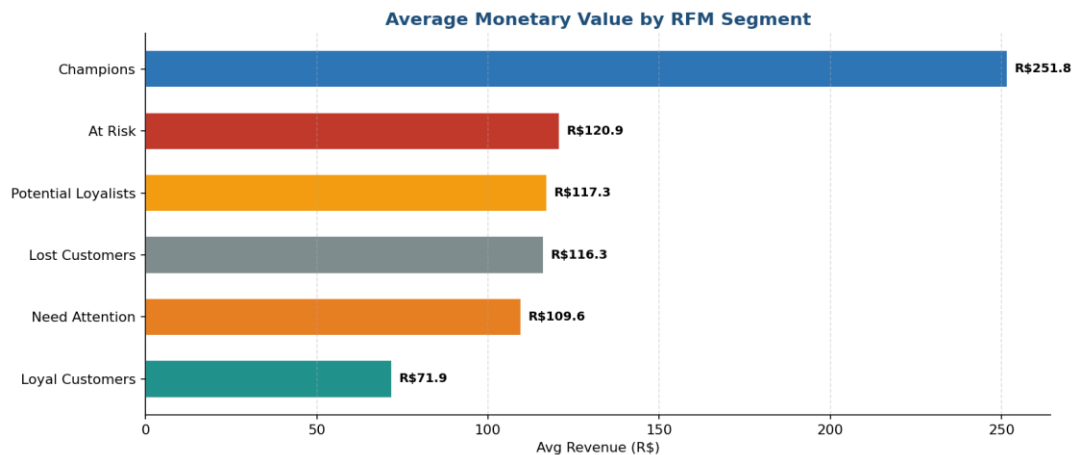


Figure 14: Champions generate R\$257.80 per customer on average — 4x more than Loyal Customers (R\$62.13) and 17x more than Lost Customers (R\$14.97).

8.5 Average Recency by Segment

Figure 15. Average Recency by RFM Segment

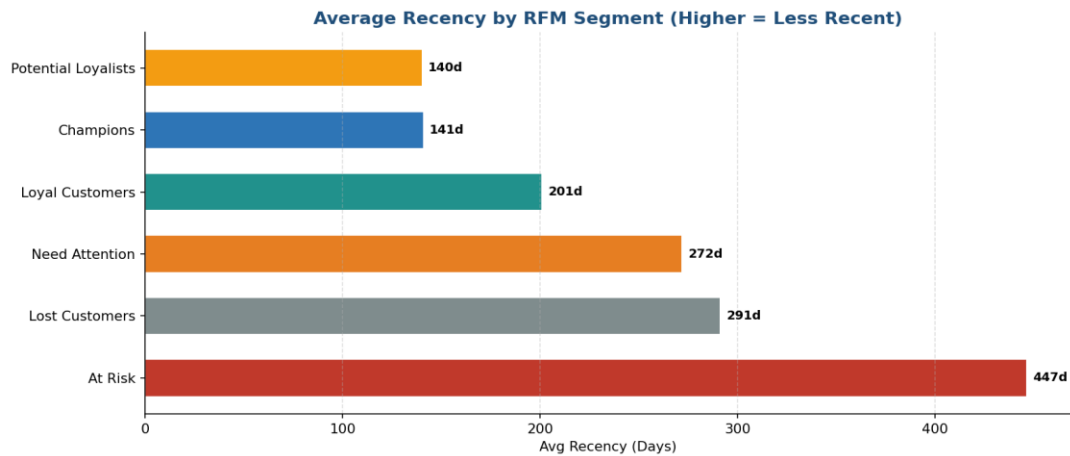


Figure 15: Lost Customers (424 days) and At Risk (421 days) have not purchased in over a year on average. Champions and Potential Loyalists are the most recent at ~114 days.

9. Segment Profiles and Business Interpretation

Table 12. Segment Profiles and Recommended Actions

Segment	Profile	Key Characteristic	Priority Action
Champions (429)	Best customers: recent, frequent, high spenders	Avg R\$257.80, 114-day recency	Reward with VIP loyalty programmes and early access
Loyal Customers (1,098)	Frequent buyers with moderate spend	Avg R\$62.13, 170-day recency	Upsell and cross-sell to increase basket size
Potential Loyalists (981)	Recent buyers, not yet frequent	Avg R\$111.25, 114-day recency	Nurture with personalised recommendations to drive 2nd purchase
Need Attention (1,870)	Above-average spend but becoming inactive	Avg R\$130.96, 266-day recency	Re-engage with targeted offers within 30 days
At Risk (1,492)	Used to purchase but not recently active	Avg R\$107.26, 421-day recency	Win-back campaigns with strong discount incentives
Lost Customers (393)	Low spend, last purchase over a year ago	Avg R\$14.97, 424-day recency	Low-cost reactivation only; suppression from heavy spend

10. Key Findings Summary

Table 13. Prioritised Key Findings

#	Finding Area	Finding	Business Impact
1	Customer Segments	Need Attention (1,870) and At Risk (1,492) together = 53.6% of all customers; urgent retention risk	Very High
2	Champions Value	Champions (only 6.9% of customers) generate 4x the avg monetary value vs Loyal Customers (R\$257 vs R\$62)	Very High
3	Geographic Dominance	SP accounts for 42.6% of all platform orders (2,670 out of 6,263 unique-customer orders)	High
4	State Recency Gap	MG (1 day) vs RR (199 days) — 198-day gap between most and least recently active states	High
5	Low Frequency Platform	Maximum customer order frequency is 3; nearly all customers order only once — structural retention challenge	High
6	Lost Customer Revenue	Lost Customers avg only R\$14.97 — already disengaged; recovery cost likely exceeds expected revenue	Moderate
7	Northern States	RR, AP, AM have near-zero frequency, recency, and monetary across all three RFM dimensions	Moderate

11. Business Recommendations

Table 14. Prioritised Recommendations

Area	Recommendation	Priority
Need Attention Re-engagement	Launch targeted re-engagement campaigns for the 1,870 Need Attention customers within 30 days. These customers have R\$130 average value and 266-day average recency — they are slipping but recoverable.	Very High
At Risk Win-Back	Deploy win-back campaigns with meaningful discount incentives for the 1,492 At Risk customers (421-day avg recency). Use email and push channels. Expected recovery rate 10–20%.	Very High
Champions VIP Programme	Protect the 429 Champions with a dedicated loyalty tier: free shipping, early product access, and exclusive discounts. These customers generate R\$257 avg revenue — even 10% churn represents significant loss.	High
Potential Loyalist Conversion	Nurture the 981 Potential Loyalists (114-day recency, R\$111 avg value) with personalised second-purchase recommendations within 45 days of their first order.	High
SP Market Deepening	SP accounts for 42.6% of orders. Invest in delivery speed, category expansion, and seller quality specifically in SP to defend and grow this dominant market.	High

MG/SC Growth	MG (most recent state, 1-day recency) and SC (8-day recency) show strong momentum. Increase campaign intensity in these markets to capitalise on active engagement.	Moderate
Lost Customer Suppression	Suppress Lost Customers (R\$14.97 avg value) from high-cost paid campaigns. Use only low-cost channels (email) for occasional reactivation attempts with minimal budget allocation.	Low

12. Limitations and Next Steps

The following limitations apply to this RFM analysis:

- This dataset contains 7,214 rows — a subset of the full 82,365-row master dataset. Segment proportions may differ when applied to the full customer base.
- RFM scoring uses equal-count quantile binning (pd.qcut). This means each score band contains an equal number of customers, which may not reflect true business value thresholds.
- The maximum observed order frequency is 3 per customer, making frequency differentiation limited. A broader observation window would produce more meaningful frequency scores.
- Segment assignment rules are threshold-based heuristics. Alternative approaches (K-means clustering, BTYD models) may produce different but complementary segmentation results.
- Monetary value includes freight charges; pure product revenue analysis would require separating price and freight_value at the line-item level.
- No demographic data (age, gender, income) is available, so segments cannot be cross-referenced with customer profiles.

Recommended next steps include: predictive churn scoring for At Risk and Need Attention segments, CLTV forecasting using BG/NBD and Gamma-Gamma models, A/B testing of re-engagement campaign designs per segment, and integration of product category data to identify which categories drive Champions vs Lost Customer behaviour.

Appendix A. RFM Scoring Rules

Table 15. Segment Assignment Rules

Segment	R Score	F Score	M Score
Champions	≥ 4	≥ 4	≥ 4
Loyal Customers	≥ 3	≥ 4	Any
Potential Loyalists	≥ 4	≥ 2	Any
Need Attention	≥ 3	≥ 2	Any
At Risk	≤ 2	≥ 2	Any
Lost Customers	Any remaining	Any remaining	Any remaining

Table 16. Analysis Outputs Generated

Output	Description
RFM_Customer_Segmentation.xlsx	Full customer-level RFM scores and segment labels exported to Excel
State Recency Table	Recency days and levels per state; Top 10 and Bottom 10 ranked
State Frequency Table	Order count and levels per state; Top 10 and Bottom 10 ranked
State Monetary Table	Revenue and levels per state; Top 10 and Bottom 10 ranked
RFM Score Table	CustomerID, R/F/M scores, RFM_Score string, and Segment for all 6,263 customers
Segment Summary Table	Customer count, avg recency, avg frequency, avg monetary by segment

*All analyses performed using Python (pandas, numpy, matplotlib, seaborn) in a Jupyter Notebook environment.
Dataset: analysis_data.csv (Olist cleaned master dataset, 7,214 rows).*